ASMI BASNET

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PROFESSIONAL SUMMARY

AI Research Engineer (Master of Science in Analytics candidate, Saint Louis University; B.E. in Electronics & Communication) with 3+ years of experience researching, building, and evaluating production Generative AI architectures, autonomous multi-agent workflows, and governed data systems. Hands-on expertise engineering sovereign, air-gapped retrieval-augmented generation (RAG) pipelines, tool-calling agent frameworks (Model Context Protocol / WebMCP, Agno), and local LLM inference architectures (Ollama: Qwen 2.5, Llama 3) with zero external data leakage. Specialized in rigorous model evaluation, chain-of-thought citation verification, and automated data quality assurance suites (IV&V) for mission-critical enterprise environments. Adept at translating complex analytical research into scalable, containerized software microservices in Python, C/C++, and SQL.

TECHNICAL SKILLS & CORE COMPETENCIES

AI Research & GenAI: Autonomous AI Agents (Agno, ReAct / Supervisor Patterns), Model Context Protocol (WebMCP Tools), Retrieval-Augmented Generation (RAG), Local LLM Inference (Ollama: Qwen 2.5, Llama 3, Phi-3), OpenAI GPT-4o API, Anthropic Claude API (Prompt Caching & Tools), Vector Search (ChromaDB, FAISS), Citation Verification, Hallucination Guardrails.
Software & Systems: Python (FastAPI, Flask, Requests, Pandas, NumPy, Scikit-Learn, PyTorch, SQLAlchemy), JavaScript / TypeScript, C/C++, Go (Golang), SQL (PostgreSQL, DuckDB, Snowflake, SQL Server: CTEs, Window Functions), RESTful APIs, Webhooks, Linux CLI, Git / GitHub, Docker, Docker Compose, CI/CD Actions.
Machine Learning & Evals: Deep Learning (PyTorch CNNs, LSTM), Supervised ML (Scikit-Learn, XGBoost, LightGBM, Random Forest, Lasso), Time-Series Forecasting (Prophet, ARIMA), Anomaly Detection (Statistical Outlier Flagging), Model Evaluation (Precision, Recall, ROC-AUC, F1, Cosine Drift).
Data Governance & Quality: Air-Gapped & Sovereign AI Architecture, Automated Data Quality Assurance (IV&V), Schema Drift Monitoring, Structured Audit Logging (JSONL), Star-Schema Dimensional Modeling, Business Intelligence (Microsoft Power BI, DAX, Tableau).

PROFESSIONAL EXPERIENCE

Saint Louis University — **St. Louis, MO**
Graduate AI Research Assistant (SPS-Corps) — *August 2025 – Present*

- Architected and deployed an air-gapped, privacy-first offline RAG compliance application in Python (FastAPI, ChromaDB, Ollama) that runs local quantized LLMs to process sensitive enterprise documentation with zero external data exposure and zero data leakage.
- Engineered a chain-of-thought verification engine evaluating retrieved context sentence-by-sentence, flagging contradictory requirements (FLAG_CONFLICT), and requesting missing context (CLARIFY) to automate governance audits.
- Designed cooperating multi-agent automation workflows leveraging the Agno framework and Anthropic Claude / OpenAI APIs, integrating custom tool-calling routines and REST API connectors that automated cross-departmental research and data synthesis, cutting manual compilation cycles by 90%.
- Engineered automated data ingestion, validation, and ETL pipelines across 8+ operational databases using Python and modular SQL CTEs with window functions, standardizing data dictionaries and cutting query execution times by 40%.
- Built a hybrid forecasting and anomaly detection engine pairing Facebook Prophet with PyTorch LSTM neural networks, automatically flagging demand surges and budget variances beyond ±2.5 standard deviations to deliver predictive resource allocation insights.
- Collaborated directly with faculty researchers and administrative leadership in Agile sprints to define acceptance criteria, author technical runbooks, and conduct hands-on user training workshops for 25+ staff members.

Saint Louis University — **St. Louis, MO**
Data & AI Solutions Analyst Intern — *June 2024 – August 2024*

- Built and deployed 6 interactive executive dashboards across Power BI and Tableau backed by star-schema data models, consolidating 15+ operational KPIs to enable leadership to track resource utilization, productivity, and project milestones.
- Engineered automated data profiling, cleaning, and validation scripts in Python (Pandas) and SQL across 10,000+ institutional records, establishing automated webhook alerts that instantly flag schema drift and transactional anomalies.
- Enforced strict institutional data governance and privacy rules, building row-level data quality validation checks across reporting layers.

Entegra Sources Pvt. Ltd. (Mayahold Group) — **Remote / Kathmandu, Nepal**
Systems & Data Analyst — *December 2022 – July 2023*

- Extracted, cleaned, and normalized multi-source operational and telemetry datasets (5,000+ records) into standardized 3NF relational schemas using Python (Pandas, SQLAlchemy) and SQL, ensuring complete data consistency across enterprise systems.
- Validated algorithmic logic and automated backend processing scripts in Python and C/C++, implementing robust exception handling, boundary checks, and audit logging to maintain 99.8% pipeline reliability.
- Authored formal technical documentation detailing data dictionaries, data validation rules, and sampling methodologies to establish organizational data governance standards.

FEATURED AI ENGINEERING & RESEARCH PROJECTS

PolicyGuard — Sovereign Offline RAG & Document Intelligence Suite — **Python, Ollama, ChromaDB, FastAPI, Docker**

- Engineered an air-gapped, sovereign RAG application running entirely on local quantized LLMs via Ollama to analyze confidential enterprise policy documents and contracts with zero third-party API exposure.
- Implemented automated chain-of-thought citation verification and dynamic cosine similarity guardrails, evaluating retrieved context sentence-by-sentence to detect contradictory clauses (FLAG_CONFLICT) and eliminate manual review overhead.

Nepal Disaster Watch — WebMCP Real-Time Autonomous Command Center — **TypeScript, WebMCP, REST APIs, GeoJSON**

- Architected an AI-driven crisis intelligence command center featuring 11 native WebMCP tools (6 read, 5 write) for OpenAI's WebMCP Challenge, synchronizing 6 live institutional feeds across all 77 districts of Nepal.
- Engineered resilient feed status adapters (live, snapshot, unreachable), allowing autonomous AI agents to dynamically query spatial risks, pan map layers, and prevent silent system failures during critical operations.

Autonomous Multi-Agent Workflow & Research System — **Python, Agno Framework, OpenAI GPT-4o / Claude API, Zapier**

- Built a cooperating multi-agent workflow pairing a Market Scout Agent with an Analytical Synthesis Agent to autonomously monitor data feeds, call external APIs, evaluate valuation anomalies, and synthesize structured executive briefs.
- Integrated event-driven Zapier webhooks and automated messaging endpoints to ingest real-time operational triggers and distribute structured summaries, reducing manual information gathering and reporting cycles by 90%.

EDUCATION & CERTIFICATIONS

Saint Louis University | Master of Science in Analytics (STEM) — **St. Louis, MO | Expected May 2026**
Tribhuvan University | Bachelor of Engineering in Electronics & Communication (STEM) — **Kathmandu, Nepal | Conferred 2023**
Certifications: Building with Claude & Anthropic API (Prompt Caching & Tools) · AI Engineer for Data Scientists Associate (DataCamp: PyTorch & Scikit-Learn) · Microsoft Certified: Power BI Data Analyst Associate (PL-300) · Project Management & Agile Certification (GlobalShala & SLU).